

Supervisor briefing — routing-policy selection study

21 September 2026 · What changed, why, and what we can now claim

1. The problem we set out to solve. When should an intelligent transportation system use one established routing policy rather than another, and can that choice be learned and evaluated as a *decision* rather than merely predicted?

2. What the earlier evidence said. The 270-run pilot found dynamic travel-time routing (DTT) best on both criteria in all 30 contexts — zero adaptive headroom, no study. The 432-run development matrix on a richer 12-signal network was built to create headroom. It did, but the *environmental* part of the design failed: the eco-routing estimator passed aggregate prediction (WAPE 14.71%) yet failed route ordering (21.62% selection loss, best route found in 3 of 8 probe panels), and ECO turned out to be outcome-identical to shortest-path routing in 21 of 24 contexts. The earlier conclusion was that the portfolio was insufficient and the study should be withheld.

3. What changed, and why. That conclusion was too broad. It correctly diagnosed the *environmental* axis as degenerate, but the *policy-selection* problem was intact underneath it. Re-analysing the same 432 runs showed shortest-path routing preferred in all eight low-demand contexts and DTT preferred in all eight high-demand contexts — a genuine, strong context-to-policy relationship that nobody had looked for, because attention was on the time-CO₂ trade-off that does not exist here. We therefore dropped ECO from the portfolio on evidence, kept it as an audited negative control, and rebuilt the study around demand-regime structure. **No new simulations were run and no existing result was discarded or reinterpreted.**

4. What we can now claim, from completed evidence.

- **Policy preference is regime-structured.** The preference between the two main policies reverses with demand, monotonically, in all eight restriction/signal cells.
- **The signal regime moves the reversal.** Under balanced signal timing it occurs between 720 and 1200 veh/h; under arterial-priority timing, between 1200 and 1680 veh/h. Signal policy, not demand alone, sets where adaptation starts to pay.
- **Adaptation is worth something and is recoverable.** Available headroom against the best fixed policy is 0.03018 mean regret units. A depth-3 regression tree, evaluated leave-one-context-out with scales and model refit per fold, recovers **79.9%** of it (regret 0.00606) and is optimal in 21 of 24 held-out contexts.
- **There is a hard boundary.** Hold out an entire demand level and the selector becomes *worse* than the fixed policy (0.07649 vs 0.03018). The requirement is regime coverage, not model capacity — a single split on demand already captures 53%.

5. The two negative controls we kept. ECO was removed from the portfolio because the environment gave it no independent environmental decision dimension — this is reported, not hidden. The multi-criteria preference layer is nearly inert here: time and CO₂ correlate at 0.982, and across the full weight range from pure-time to pure-emissions *not one* of the 24 selector decisions changes. Both controls are what stop the 79.9% figure being over-read as a rich multi-objective result.

6. What we deliberately did not do. We did not repair the emissions observer (not needed once ECO left the portfolio), did not rebuild the network, did not run the planned 2,800-run experiment, and did not fit any model more complex than a shallow tree. We also did not claim novelty: a targeted literature audit found demand-dependent routing benefit already established for re-routing decisions, so the paper states differences from named prior studies rather than asserting a first.

7. The one open item. The demand grid is 480 veh/h coarse, so the reversal is *bracketed, not localised*. Localising it needs two extra demand levels (960 and 1440 veh/h) over the existing cells: **144 runs, about 1.5 minutes of compute**. This was designed and scripted but **not run**, because the SUMO network and configuration files were not in the material supplied — a reconstructed network would not be comparable with the 432 completed runs. The paper reports brackets and makes no threshold estimate, so nothing in it depends on this experiment.

8. Contribution statement. An empirical characterisation of demand- and signal-regime-dependent routing-policy preference in a controlled microsimulation; quantification of the decision headroom that policy switching creates; a held-out, decision-focused evaluation showing an interpretable selector recovers most of that headroom; and identification of an extrapolation boundary beyond which adaptive selection is harmful. No new routing algorithm, no new multi-criteria method, no new learning algorithm.

Status: manuscript, figures, tables, bibliography and analysis scripts complete; every reported number regenerates from the exported run table. Outstanding decisions for you: conference template/page limit (none was supplied — currently generic A4) and the author block.